Scheme and answer key for

QP CODE: 25900559

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY201 - COMPUTATIONAL PHYSICS-C++ PROGRAMMING (2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A: Short Answer Type Questions

Answer **any 10** questions Each question carries **2 marks**

Q No.	Question	Suggested Answer and Marking Breakup	Marks
1.	Why are << and >> operators used in C++ programming?	The **insertion** operator (<<) is used for **output** (inserting data into a stream, e.g., cout). (1 Mark) The **extraction** operator (>>) is used for **input** (extracting data from a stream, e.g., cin). (1 Mark)	2
2.	Define keywords and give four examples of C++ keywords.	**Definition:** Reserved words in C++ that have a predefined meaning and cannot be used as identifiers (1 Mark) **Examples (Any four):** int, float, if, else, for, while, class, return, void, etc. (4 x 0.25 = 1 Marks)	2
3.	From the given list (C, C++, FORTRAN, Python, COBOL, Java) Classify the languages into object-oriented and procedure-oriented categories.	**Object-Oriented (OOP): C++, Python, Java**. (1 Mark)] **Procedure-Oriented (POP): C, FORTRAN, COBOL**. (1 Mark) (reduce marks accordingly for the mismatch)	2

4.	Write a for loop to print numbers from 10 down to 1.	Correct for loop structure with correct syntax:
		for (int i = 10; i >= 1; i--)
		{
		cout << i << endl;
		}
		Breakup: Initialization (i=10): 0.5 Mark(M), Condition (i>=1): 0.5 M, Update (i--): 0.5 M, Body (printing): 0.5 M.
			2

Q No.	Question	Suggested Answer and Marking Breakup	Marks
5.	Write the difference between break and continue statements in C++.	**break:** Terminates the innermost loop (or switch statement) and transfers control to the statement following the loop/switch. (1 Mark) **continue:** Skips the rest of the current iteration of the loop and proceeds to the next iteration. (1 Mark)	2
6.	What is the function of the parameter list in C++?	Its function is to pass **data/information** into the function for its operation/computation. (2 Mark)	2
7.	What are one-dimensional arrays?	A collection of a fixed number of elements of the **same data type**. (1 Mark) Stored in **contiguous memory locations** and accessed using a **single index**. (1 Mark)	2
8.	Mention any two category of functions with examples.	**Category 1 Standard Library Functions** (sqrt(), cin(), cout()) (1 Mark) **Category 2 User-Defined Functions** (int add(...), void print(...)). (1 Mark) Or **Functions in C++ can be classified in multiple ways, including by their origin, return type, and parameter** **Return Type**: `void` vs `int` functions **Parameter Usage**: With or without parameters b**ehavior**: Recursive, inline, overloaded **therefore, any correct classification with example can give marks**	2
9.	What values are given inside the brackets of a function call?	They are called **Arguments** (or Actual Parameters). (1 Mark) They are the **specific values** that are passed to the function's corresponding formal parameters. (1 Mark)	2
10.	State the advantages of using operator overloading in C++.	**Readability:** **Consistency:** **Simpler Syntax:**	2

Q No.	Question	Suggested Answer and Marking Breakup	Marks
		Flexible Semantics: **Operator Reuse:** . **Better Modeling:** (Explain **Any two valid points:** 2 Mark)	
11.	What do you mean by an array containing objects of a class?	It is an array where each element is an **object (instance)** of a specified class. (1 Mark) Used to handle a **group/collection of objects** of the same class conveniently. (1 Mark)	2
12.	What are the benefits of making an array a member of a class?	Array data is **encapsulated** along with other class members Access and manipulation are restricted to the class's **member functions**, protecting the data The array is intrinsically **tied to the object's identity** and purpose, improving class design. The array is **created and destroyed automatically** when the class object is instantiated or destroyed. The array becomes a simple part of the **overall object state**, making the object easier to manage as a single unit. **(Any two valid points:** 2 marks)	2
13.	What do you mean by a pointer in C++ programming?	A pointer is a variable that stores the **memory address** of another variable. (1 Mark) Used for dynamic memory management, accessing arrays, and implementing data structures (1 Mark)	2
14.	Explain the concept of static member functions in a class.	A static member function **belongs to the class** as a whole, not to a specific object. (1 Mark) **Characteristic (Any one):** Can be called **without creating an object**, or can only access static member data. (1 Mark)	2

Part B: Short Essay Type Questions

Answer **any 6** questions Each question carries **5 marks**

Q No.	Question	Suggested Answer and Marking Breakup	Marks
15.	Create an algorithm that determines which of three numbers is the largest.	**Algorithm Steps:** 1. Start, Input three numbers (A, B, C). (1 Mark) 2. IF (A > B) AND (A > C) THEN Output "A is largest." (2 Marks) 3. ELSE IF (B > C) THEN Output "B is largest." (1 Mark) 4. ELSE Output "C is largest." (1 Mark) 5. End.	5
16.	Write a C++ program to demonstrate logical AND, OR, and NOT operations.	Any similar program demonstrating logic operators **Program:** **#include <iostream>** **int main() {** **bool a = true;** **bool b = false;** **// Output is 1 for true, 0 for false.** **std::cout << "AND: " << (a && b) << "\n";** **std::cout << "OR: " << (a || b) << "\n";** **std::cout << "NOT: " << (!a) << "\n";** **return 0;** **}** **Output:** AND: 0 OR: 1 NOT: 0 (program structure :2 marks, , use of operator for logic gates AND, OR and NOT= 3marks)	5

17.	Define a nested loop in C++ and demonstrate its working using a simple program.	**Definition:** A loop that is placed inside the body of another loop. The inner loop executes completely for every single iteration of the outer loop. (2 Marks)
		
		Program/Demonstration: (Any similar program with nested loop)
		#include <iostream>
		
		int main() {
		// Outer loop (i) controls the row number
		for (int i = 1; i <= 2; i++) {
			5

Q No.	Question	Suggested Answer and Marking Breakup	Marks

		// Inner loop (j) controls the column number
		for (int j = 1; j <= 2; j++) {
		std::cout << j << " "; // Print column number
		}
		std::cout << "\n"; // Newline after each row
		}
		return 0;
		}
		(program structure :1 mark, use of nested loop 2 marks)	

18.	(a) What is the purpose of an if statement in C++? (b) Write the basic syntax of an if statement in C++? (c) What data type must be in the condition of an if statement?	**(a) Purpose:** Used for **conditional execution**; it executes a block of code *only if* a specified condition is true. (2 Marks)
		(b) Basic Syntax:
		if (condition)
		{
		statement(s);
		} (2 Marks)
		(c) Data Type: Must evaluate to a **boolean** value (true or false). (1 Mark)
			5

19.	Compare switch statements and if...else-if constructs in C++ and highlight the differences.	Can be classified based on 1. **Condition Type:** switch tests a single variable vs. constant integer/char values; if-else-if tests multiple complex boolean expressions. 2. **Operator Use:** switch uses only equality (= =); if-else-if can use any relational/logical operator (<, >, &&, etc.). 3. **Range Testing:** switch cannot easily check ranges; if-else-if is ideal for range checks. 4. **Flexibility/Efficiency:** if-else-if is more flexible; switch is often more efficient for many specific constant checks. 5. **Flow Control:** if-else-if code blocks are executed exclusively; no fall-through mechanism.switch Requires the break statement to exit a case and prevent fall-through (any three points give 5 marks, 2 points 3.5 marks, 1 point- 2 marks)	5

20.	Show how to declare and initialize a 2D array in C++ with an example.	**Declaration/Syntax:**
		data_type arrayName[row_size][column_size];
		(2 Marks)
		Example
		int matrix[2][3] = {
		{10, 20, 30},
		{40, 50, 60}
		};
		(3 Marks)
			5

Q No.	Question	Suggested Answer and Marking Breakup	Marks
21.	What are the essential elements of a function in C++? Describe each briefly.	Describe the following 5 elements briefly 1. **Return Type** (Data type of the value returned, or void) (1 Mark) 2. **Function Name** (Unique identifier for calling) (1 Mark) 3. **Parameter List** (Input values/arguments enclosed in ()) (1 Mark) 4. **Function Body** (Statements defining function's action, enclosed in {}) (1 Mark) 5. **Return Statement** (Optional; sends value back to caller) (1 Mark) **or** 1. **Declaration** Tells the compiler about the function's name, return type, and parameters. Example: int add(int a, int b); (2 Marks) 2. **Definition** Provides the actual body of the function with the logic to execute. Example: int add(int a, int b) { return a + b; } (1.5 marks) 3. **Call** Executes the function using actual arguments. Example: int result = add(5, 3); (1.5 marks)	5
22.	Describe the steps involved in opening, reading, writing, and closing a file in C++.	**Steps using <fstream>:** 1. Include Header (<fstream>) (0.5 Marks) 2. Create a Stream Object (ofstream, ifstream, or fstream) (1 Mark) 3. **Opening:** Use the open() function with filename and mode (ios::in/ios::out). (1.5 Marks) 4. **Reading/Writing:** Use >> (extraction) or << (insertion) operators with the stream object. (1 Mark) 5. **Closing:** Use the close() member function. (1 Mark)	5